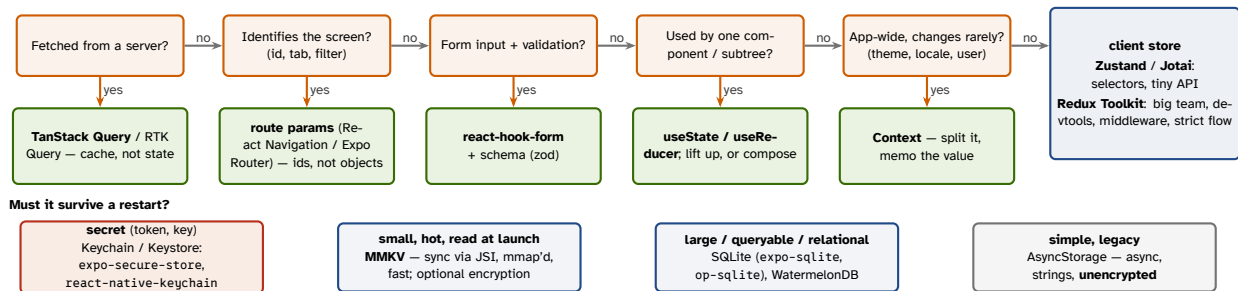


**In one line:** “Which state library?” is the wrong question. Classify first — **server, navigation, local UI, shared client, form, persisted** (*secret?*) — each has a best-fit tool; most bugs are one kind kept in another’s tool.

Decide per piece of state — first “yes” wins:



## Server state — TanStack Query (v5)

- Cache keyed by queryKey ([ 'todo', id ]); dedupes requests; **stale-while-revalidate**: show cached, refetch behind.
- staleTime (default 0) — how long data is fresh (no refetch on mount/focus). gcTime (default 5 min; v4 cacheTime) — how long an *unused* entry stays in memory.
- invalidateQueries after a mutation: mark stale, refetch active. retry: 3, exponential backoff.
- RN wiring: focusManager ← AppState, onlineManager ← NetInfo. Offline, queries pause; persist the cache with PersistQueryClientProvider.
- RTK Query: the same inside Redux — createApi, providesTags / invalidatesTags.

## Example — optimistic update with rollback

```

const qc = useQueryClient();
const save = useMutation({
  mutationFn: (t: Todo) => api.update(t),
  onMutate: async (t) => {
    await qc.cancelQueries({ queryKey: ['todos'] }); // no race
    const prev = qc.getQueryData<Todo[]>(['todos']);
    qc.setQueryData<Todo[]>(['todos'], (old = []) =>
      old.map((x) => (x.id === t.id ? t : x)));
    return { prev }; // -> ctx
  },
  onError: (_, t, ctx) => qc.setQueryData(['todos'], ctx?.prev),
  onSettled: () => qc.invalidateQueries({ queryKey: ['todos'] });
});

```

## Context vs stores — re-render mechanics

- Context** re-renders *every* consumer when the value identity changes — no selectors. Low-frequency values only; split state from dispatch, useMemo the value.
- Redux Toolkit**: createSlice (reducers “mutate” a draft, Immer makes the immutable copy), configureStore, createAsyncThunk. useSelector re-renders when its result changes by reference; derive with memoised createSelector.
- Zustand**: create(fn), no provider; useStore(selector) subscribes to one slice; object selectors need useShallow; persist middleware.
- Jotai**: atoms, bottom-up; derived atoms re-run only for their readers.

## Forms

**react-hook-form**: values live outside React state — a keystroke does not re-render the form; RN inputs go through Controller; validate with a zod resolver.

## Persistence — beyond the picture

- AsyncStorage on Android is SQLite with a default size cap (*unverified*); MMKV’s sync reads let settings load at launch without a spinner — its encryption key belongs in the Keychain. Secure stores hold *small* values (tokens), not data.
- Persist client state selectively and **version + migrate** its shape.

## Offline-first & normalisation

- Local DB is the source of truth; the UI reads it; sync is background.
- Writes go to an **outbox**, replayed on reconnect with **idempotency keys** (a retried POST must not double-charge).
- Conflict rule per entity: last-write-wins, server-wins, or field merge.
- Normalise** shared entities ({ids, entities}, RTK createEntityAdapter): one update reaches every screen. Query caches are per key, *not* normalised — setQueryData or invalidate.

## Traps

- ! Copying fetched data into Redux/useState: two sources of truth, stale forever. Server state belongs in the query cache.
- ! Tokens in AsyncStorage (plain file/SQLite) — use Keychain/Keystore.
- ! staleTime: 0 + many mounts = refetch storms; gcTime is *not* freshness.
- ! useStore() without a selector re-renders on every change.
- ! Whole objects as route params — stale copies, broken deep links.

## Remember

**Server** → **query cache** • **screen** → **route** • **local** → **useState** • **rare global** → **Context** • **hot shared** → **store** • **secret** → **Keychain**.